Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/20/2006
PI Name:	Limoli, Charles Ph.D.		
Project Title:	Mechanisms Of High LET Radiation Induced Genomic Instability In The CNS		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	03/01/2007	**End Date:**	02/28/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Start date changed for FY2007 to be 3/1/2007 as PI's anniversary date (per A. Chu 12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA06CB37G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space radiation environment poses a range of potential health risks to astronauts. Exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can damage critical cellular targets leading to a number of deleterious effects. Space travel has also been found to elicit changes in the redox state of cells that may exacerbate these adverse effects. These risks may be particularly problematic to stem/precursor cells within the CNS, where exposure to irradiation may promote genomic instability which could result in cancer and/or cognitive dysfunction. In vitro and in vivo we have shown neural precursor cells are exquisitely sensitive to irradiation, undergoing a dose-dependent induction of apoptosis that is associated with a chronic oxidative stress. These effects are strongly correlated with a persistent inhibition of neurogenesis that may underlie cognitive deficits observed after cranial irradiation. We have also shown that human brain tumors contain a significant number of cells that express many of the same immature lineage-specific markers common to neural precursor cells, supporting a possible role for these cells in the development of brain tumors. Based on these observations, we hypothesize that neural precursor cells exposed to high energy ions will be particularly susceptible to genomic instability and in a manner that depends upon redox state. Genomic instability is a process believed to drive cancer progression, and recently, redox state has been implicated in the induction and perpetuation of genomic instability. Redox state has been found to be a critical determinant in regulating the response of the CNS to a variety of damaging treatments, including irradiation. Changes in the redox environment are often associated with alterations in cellular metabolism which can promote oxidative stress, and these types of changes are common to tumors including those within the CNS. Because exposure to radiation has been found to cause genomic instability, and since oxidative stress is associated with the radiation response of neural precursor cells, it follows that changes in redox state might constitute a biochemical mechanism for modifying the radioresponse of the CNS. We are proposing a series of studies to test the overall hypothesis that alterations in cellular redox state promote genomic instability that can adversely impact the CNS upon exposure to high LET radiation. While the development of genomic instability may predispose neural precursor cells to transformation events in the CNS, the unstable phenotype may also lead to radioresistance that changes the phenotypic fate of neural precursor cells that impairs neurogenesis. We propose that changes in redox state play a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo; changes that will foster the onset of radiation-induced genomic instability, radioresistance and altered neurogenesis. Using an in vitro and in vivo approach the goals of our project will be addressed in the following 3 specific aims: Specific aim 1: Elucidate the dose-response relationship for the induction of genomic instability in neural precursor cells cultured in vitro or isolated from mouse brain after exposure to protons and iron ions. Specific aim 2: Determine if chronic changes in cellular redox state alter the dose response relationship determined above and predispose neural precursor cells to proton- and iron ion-induced genomic instability. Specific aim 3: Determine if differentiation of genomically unstable neural precursor cells is altered in vitro, or if neurogenesis is altered in vivo when unstable precursor cells are implanted into the brains of control or irradiated mice. : Task Objective: To determine whether changes in redox state adversely impact the CNS. We will assess whether oxidative stress alters the radiosensitivity of neural precursors and predisposes them radiation-induced genomic instability. Experiments will ascertain whether such changes alter precursor cell plasticity in a manner that compromises their normal function. These studies will promote an understanding of the space radiation effects on the CNS. Data collected from these studies will improve both our understanding of the basic biology underlying CNS radiation effects, and in assessing risks to astronauts engaged in space travel. This work will establish the extent to which redox related changes derived from environmental stressors (oxidative injury) and/or damaging agents (ionizing radiation) are problematic to astronauts and whether specific countermeasures can be developed to minimize such risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will shed light on many of the basic biological processes that are associated with exposure of the CNS to ionizing radiation. As such, our work is directly relevant to those involved in the treatment of CNS tumors via radiotherapy. Such individuals include clinicians, basic scientists and even patients, all of whom have a vested interest in minimizing side-effects associated with the exposure of normal tissue to ionizing radiation. Outside of the clinical relevance, our work should foster a general interest to those members of the public supporting the long-term goals of NASA involving the expoloration and colonization of space.
Task Progress:	We have made solid progress during the first half of our funding period. Our findings clearly substantiate a major hypothesis of our overall grant, namely, that oxidative stress has a significant impact on changes radiosensitivity. Our capability to alter the redox status of neural precursor cells through changes in cell density allows us to manipulate the microenvironment in the absence of significant cellular perturbations. This approach has allowed us to analyze how changes in oxidative stress impact radiosensitivity under conditions that are minimally toxic and easily manipulated. This methodology has revealed the dual nature of oxidative stress. While acute increases in ROS are initially mitogenic, persistent increases in oxidative stress are growth inhibitory and radiosensitizing. Additional strategies using exogenous oxidants such as hydrogen peroxide or mitochondrial electron transport chain inhibitors (antimycinA, menadione) that elevate mitochondrial-derived ROS all provide consistent results, and show that under conditions of prolonged oxidative stress, cells become more sensitive to the action of ionizing radiation. Work conducted at Loma Linda University has demonstrated qualitatively similar results where conditions used to increase oxidative stress were found to enhance the sensitivity of neural precursors to 250 MeV protons. These positive results provide the framework for progressing to those studies aimed at determining how oxidative stress impacts genomic instability, differentiation and neurogenesis in the CNS. Recent data gathered this last year also points to some intriguing similarities between primary neural precursor cells and brain tumor cells. A comparison between two glioma lines and neural precursor cells has revealed that these cells have similar post-irradiation redox responses. Both gliomas and neural precursor cells exhibited similar increases in oxidative stress following irradiation, a trend that was distinctly opposite when normal human brain cells were analyzed. Furthermore, neural precursor cells were found to exhibit similar phenotypic expression profiles as glioma cells. The foregoing evidence points to a number of shared properties between neural precursor cells and brain tumor cells, and suggests, that under the appropriate conditions, neural precursor cells may be predisposed to transformational events, representing a potentially deleterious pool of cancer stem cells in the brain. Major accomplishments are summarized below:

	1) In vitro studies using hippocampal precursors have revealed that endogenous oxidative stress increases the sensitivity of these cells to X-rays. 2) In vitro studies using hippocampal precursors have revealed that exogenous oxidative stress increases the sensitivity of these cells to X-rays and to 250 MeV protons. 3) In vitro studies have shown that perturbation of mitochondrial electron transport increases oxidative stress and enhances the sensitivity of neural precursors to irradiation. 4) Comparisons between neural precursor cells and brain tumor cells have shown that these cells share similar redox properties after irradiation. Neural precursor cells and gliomas show marked increases in oxidative stress after irradiation, an effect not found when normal brain cells were analyzed after irradiation. 5) Neural precursor cells and brain tumor cells express a number of shared phenotypic markers. The similar expression of immature markers suggest that a fraction of brain tumors contain cells similar to neural precursor cells. 6) We have isolated and expanded over a dozen telomere tagged cultures of neural precursor cells derived from individual Tel+/- mice. These cells have been tested for puromycin resistance, azacytidine resistance and are ready for irradiation and ganciclovir selection to monitor radiation-induced genomic instability. 7) We have published 4 manuscripts, with two more nearing completion by early 07.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Limoli CL, Rola R, Giedzinski E, Mantha S, Huang TT, Fike JR. "Cell-density-dependent regulation of neural precursor cell function." Proc Natl Acad Sci U S A. 2004 Nov 9;101(45):16052-7. PMID: 15522966 , Nov-2004
Articles in Peer-reviewed Journals	Limoli CL, Giedzinski E, Baure J, Rola R, Fike JR. "Altered growth and radiosensitivity in neural precursor cells subjected to oxidative stress." Int J Radiat Biol. 2006 Sep;82(9):640-7. PMID: 17050476 , Sep-2006
Articles in Peer-reviewed Journals	Limoli CL, Giedzinski E, Baure J, Rola R, Fike JR. "Redox changes induced in hippocampal precursor cells by heavy ion irradiation." Radiat Environ Biophys. 2006 Nov 14; [Epub ahead of print] PMID: 17103219 , Nov-2006